AI INTERNSHIP
WEEK 1
PROGRESS REPORT

PRESENTED BY:

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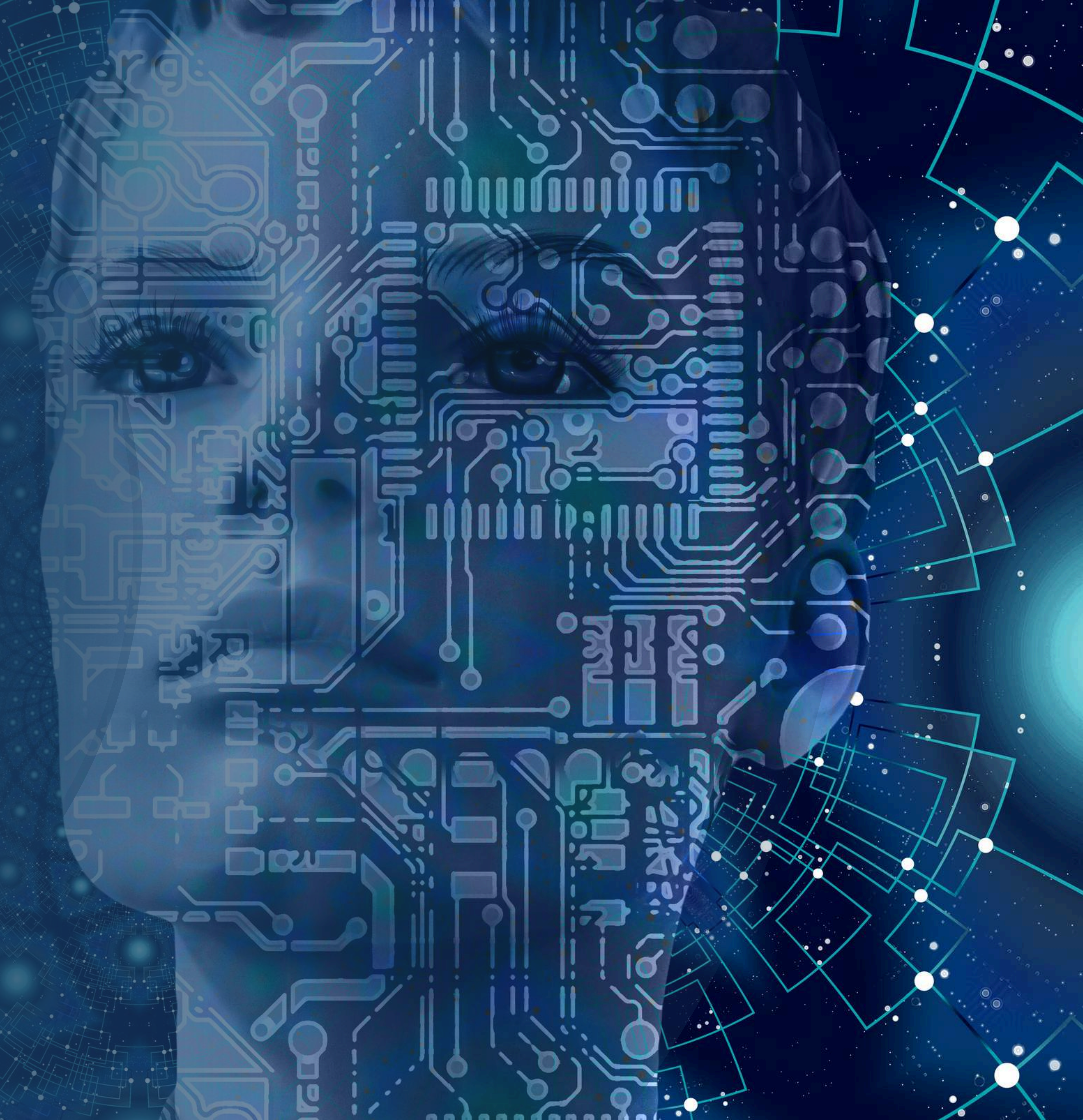
AI INTERN | XEVEN SOLUTIONS



WEEK 1 OVERVIEW

- AI Fundamentals
- Python Programming
- Conditional Statements & Logic
- Operators & Type Conversion
- Machine Learning Fundamentals
- Python Lists & Operations
- Practical Coding Tasks
- Git & GitHub

WEEK 1 LEARNING JOURNEY

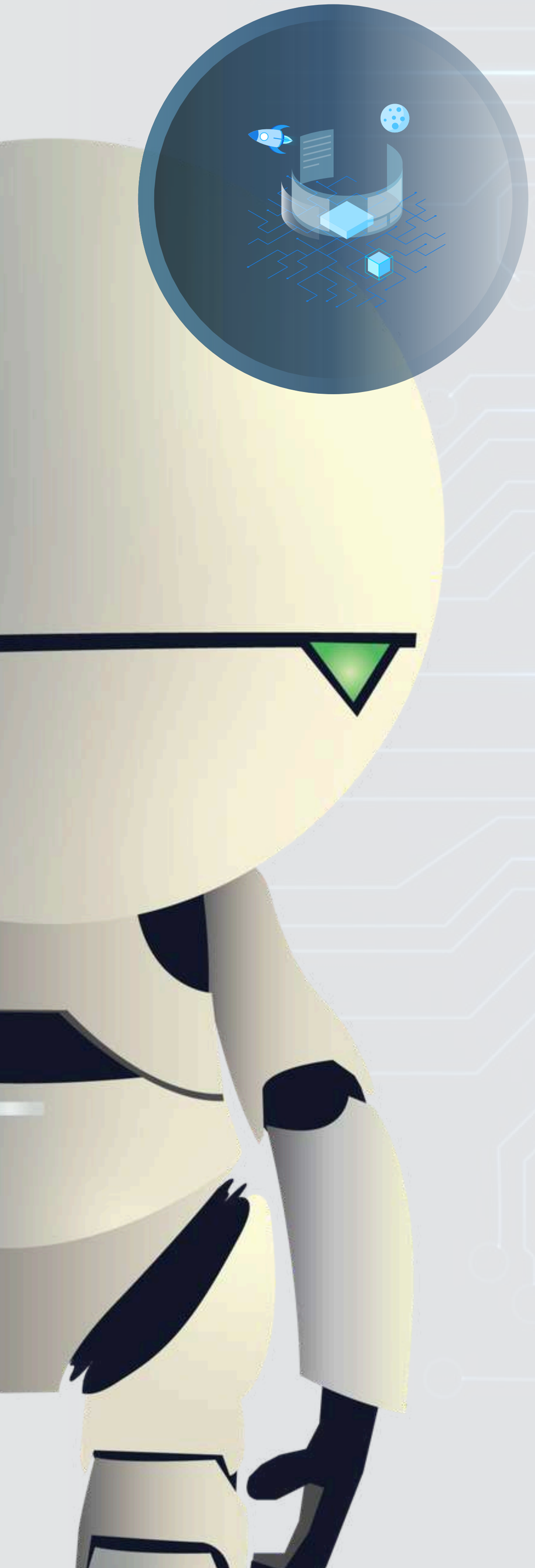
Content:

- Day 1: AI Fundamentals & Environment Setup
- Day 2: Python Data Types
- Day 3: Conditional Statements & Logic
- Day 4: Operators & Type Conversion
- Day 5: Machine Learning Fundamentals
- Day 6: Python Lists & Operations

DAY 4 - ADVANCED LOGIN SYSTEM & CALCULATOR

Content:

- User Input Validation
- Logical & Comparison Operators
- Type Conversion
- Error Handling
- Multi-Operation Calculator



DAY 4 - CODE WALKTHROUGH

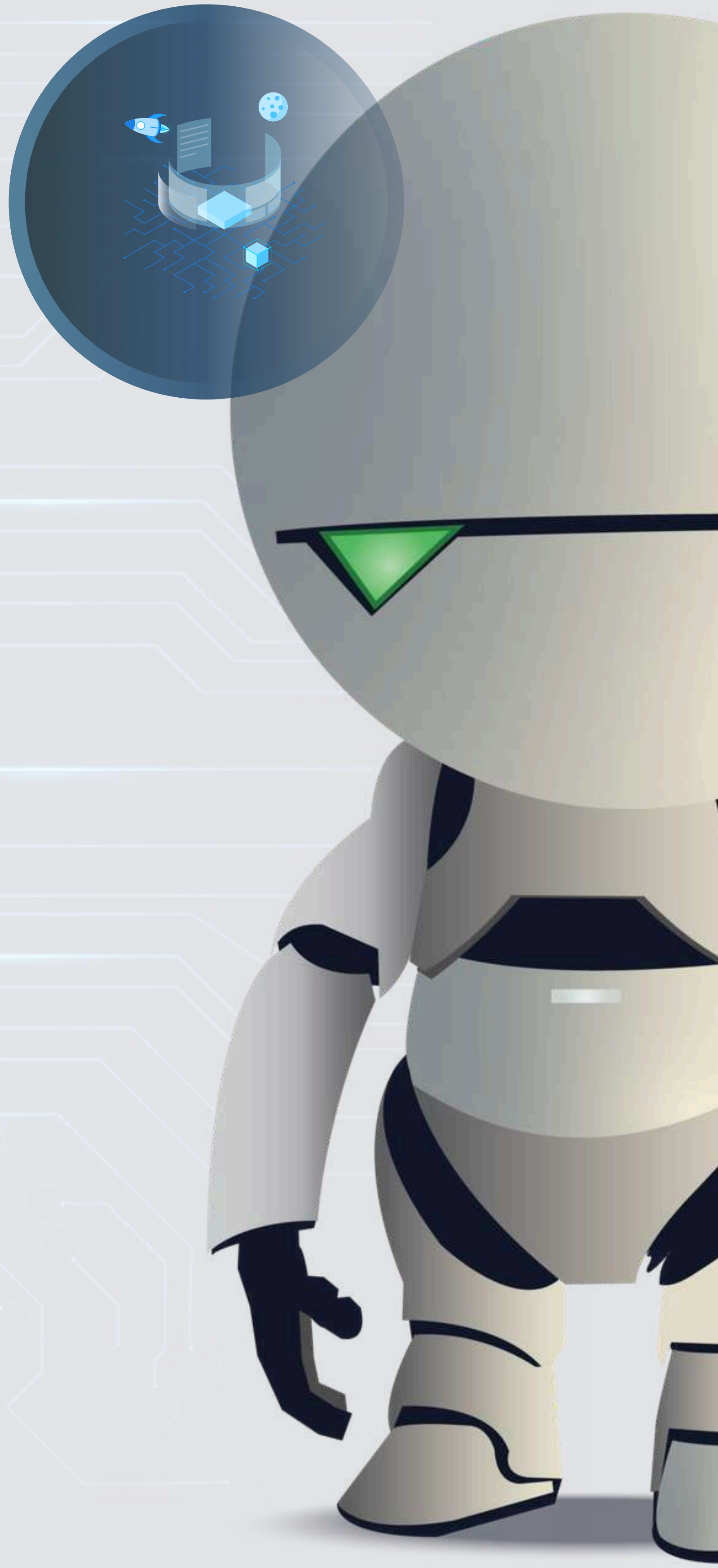
```
if len(username) < 5:
    print("Error: Username must be at least 5 characters long.")

# Validate password
if len(password) < 8:
    print("Error: Password must be at least 8 characters long.")

# Validate age
if age < 18:
    print("Error: You must be 18 or above to access the system.")
```

Short Points:

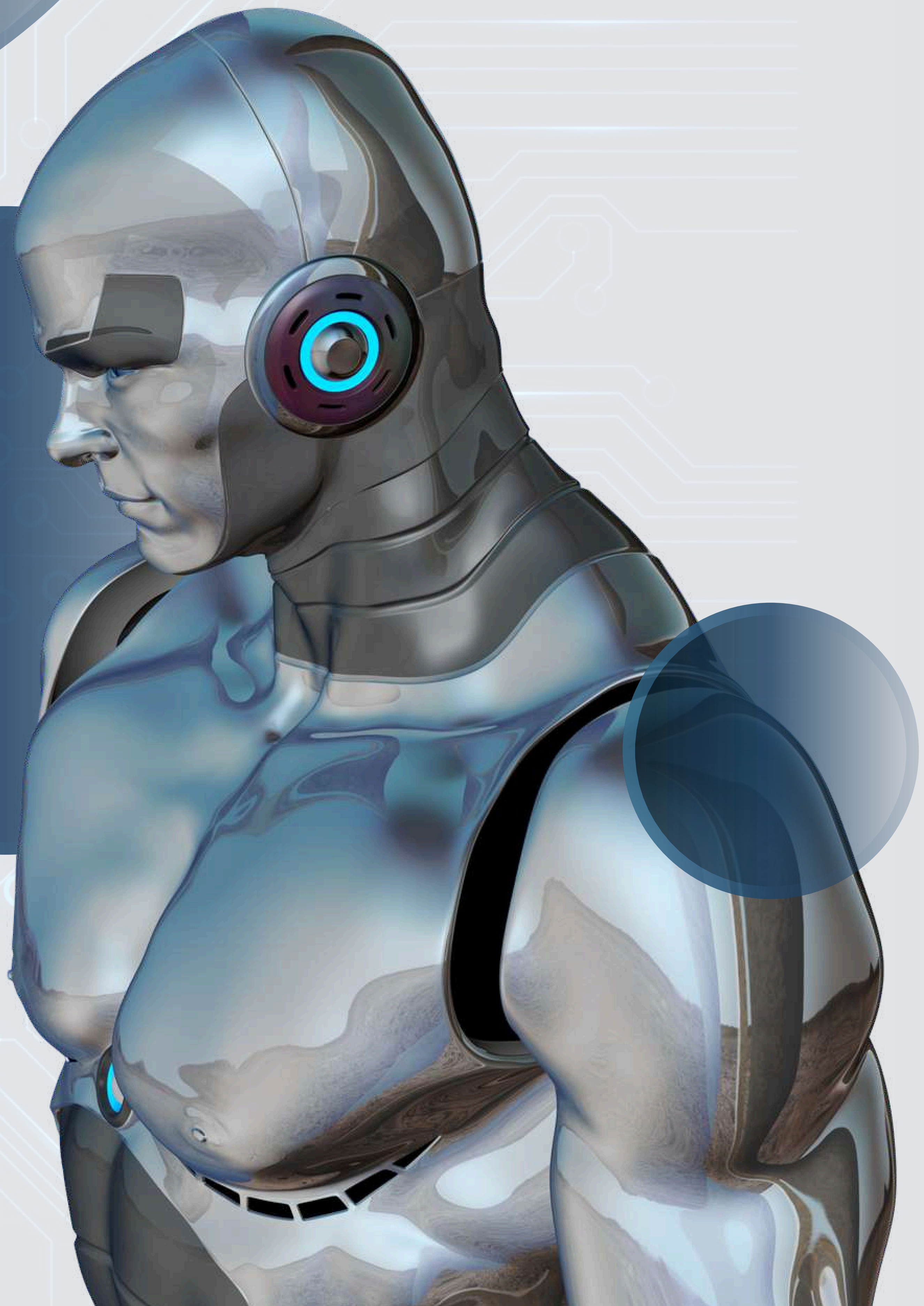
- Input validation
- Conditional logic
- Error messages



DAY 5 - MACHINE LEARNING FUNDAMENTALS

Points:

- Supervised Learning
- Unsupervised Learning
- Reinforcement Learning
- Regression
- Classification



Goal: Understanding how machines learn from data

RESEARCH PROCESS & SOURCE COMPARISON

<u>**Points:**</u>

- Researched AI/ML concepts from multiple sources
- Compared information from ChatGPT, Gemini & Claude
- Read and reviewed an external article
- Identified similarities and differences
- Synthesized the information into clear conclusions

ChatGPT → Gemini → Claude → Article

Compare → Verify → Synthesize

CHALLENGES & SOLUTIONS

Challenge 1 — Coding Errors
Solution: Debugged code and tested each part step-by-step.

Challenge 2 — Understanding Concepts
Solution: Used multiple sources and practical examples.

Challenge 3 — Git/GitHub Issues
Solution: Practiced Git commands and resolved errors through debugging.

KEY TAKEAWAY & NEXT STEPS

Key Takeaway:

Hands-on practice and debugging helped me understand technical concepts more effectively.

Week 2 Goal

Python Data Structures & Algorithms

- Lists & Dictionaries
- Loops
- Data Manipulation
- Algorithmic Thinking
- Real-World Python Systems

CONCLUSION

Points:

- Built a strong foundation in Python and AI
- Completed practical coding tasks
- Improved debugging and problem-solving skills
- Learned how to research and compare technical information
- Prepared for more advanced Python concepts in Week 2

THANK YOU.